

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct

I **Anchal Kumari(192312255)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Anchal

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Citation, Domain, and ML Problem

The article is authored by Moslehi, Rabiei, Soltanian, and Mamani, and was published in 2022 in BMC Medical Informatics and Decision Making, a peer-reviewed Springer Nature journal (DOI: 10.1186/s12911-022-01939-x). The domain is clinical healthcare analytics, specifically the classification of patient mortality risk during the COVID-19 pandemic. The core machine learning problem is a supervised binary classification task: predicting whether an admitted COVID-19 patient will recover or die, based on 25 demographic, clinical, and laboratory features recorded at the time of hospital admission.

(b) Key Objective and Research Gap

The authors set out to build a mortality-classification model for COVID-19 patients that is both highly accurate and clinically interpretable, so that physicians can quickly identify high-risk patients and prioritise treatment decisions. They identify a clear research gap: prior machine-learning studies on COVID-19 mortality (e.g., using boosting or ensemble methods) achieved strong predictive accuracy but functioned largely as black-box models, offering little insight into how individual demographic or laboratory features actually drove the prediction. Purely statistical comparative studies, on the other hand, reported associations but did not produce a deployable decision-support tool. To close this gap, the authors propose a two-stage pipeline: first, a random-forest feature-importance step to shortlist the most influential predictors from 25 candidate features, and second, a comparison of four white-box decision-tree variants (LMT, C4.5, C5.0, CART) trained on those shortlisted features. The best-performing tree is then presented as an interpretable flowchart, so that the model's reasoning — not just its prediction — is directly usable by clinicians at the bedside.

Task 2: Methodology Analysis

(a) ML Technique(s) and Dataset

The authors combine two ML techniques. First, a Random Forest (an ensemble of 1,000 trees using the Gini criterion) is used purely for feature selection — ranking all 25 candidate features by relative

importance and retaining only those above a 6% importance threshold (the third-quartile cut-point). Second, four supervised decision-tree classifiers are trained and compared on the shortlisted features: the Classification and Regression Tree (CART), C4.5, C5.0, and the Logistic Model Tree (LMT), which fits a logistic regression at each node using LogitBoost.

Dataset: The study uses retrospective records of 2,470 COVID-19 patients admitted to two hospitals (Sina and Besat) in Hamadan, Iran, between February 2020 and July 2021, confirmed positive by RT-PCR. Of these, 1,853 patients (75.02%) recovered and 617 (24.98%) died. Twenty-five candidate features were recorded, spanning demographics (age, sex, marital status, smoking, comorbidities such as diabetes, hypertension, cancer) and laboratory biomarkers (BUN, PTT, SGOT, ESR, sodium, platelets, CPK, etc.). Preparation involved removing any patient record with missing or outlier values (verified to be missing completely at random, so removal introduced no bias), followed by 10 repeated random train/test splits to reduce variance from a single split, with performance averaged across the ten repetitions.

(b) Mapping to CO4 and Strength/Limitation

This methodology maps directly onto two CO4 topics. It is a Decision Tree application, since the core classifiers (CART, C4.5, C5.0, LMT) are all tree-based inductive learners that recursively partition the feature space using impurity measures (Gini index, entropy, information-gain ratio). It also connects to Statistical Learning Methods, since the LMT variant embeds a logistic-regression model at each tree node, and the overall workflow is a form of Inductive Learning — the model generalises decision rules from 2,470 labelled training examples to classify unseen patients.

Methodological Strength: The two-stage design (Random Forest for feature selection, then interpretable trees for final classification) balances accuracy with clinical usability. Using a comparatively large sample (2,470 patients) and 10 repeated splits also reduces the risk of overfitting and gives more stable performance estimates than a single train/test split.

Methodological Limitation: The data come from only two hospitals in a single Iranian province, which limits geographic and demographic generalisability. Additionally, patients with any missing feature values were dropped entirely rather than imputed, which could bias the sample toward patients with more complete (and possibly less severe) clinical records.

Task 3: Results & Critical Evaluation

(a) Key Results

The four decision-tree models were compared using Accuracy, Recall, and F1-score. CART achieved the best overall performance and was selected as the final model; it achieved an AUC of 79.5% on the ROC curve. Key figures reported by the authors are summarised below:

Model	Dataset	Accuracy	F1-Score	Recall	AUC
LMT	Training	76.89%	85.73%	—	—
C4.5	Training	77.70%	86.08%	—	—

Model	Dataset	Accuracy	F1-Score	Recall	AUC
C5.0	Training	76.89%	85.74%	—	—
CART	Training	78.24%	86.81%	—	—
CART (best model)	Test set	78.24%	86.81%	95.50%	79.5%
CART (best model)	Total dataset	78.34%	86.67%	93.85%	79.5%

The final CART tree contained 35 nodes (16 internal nodes and 18 leaves) built on six retained features — gender, age, BUN, PTT, SGOT, and ESR. It showed, for example, that patients under 60 with $\text{BUN} \leq 20$, $\text{SGOT} \leq 45$, and $\text{ESR} \leq 30$ had only a 3.6% mortality risk, whereas patients over 60 with $\text{BUN} > 20$ and $\text{SGOT} > 45$ had a 62.3% mortality risk.

(b) Critical Evaluation

The reported metrics are realistic rather than inflated: an accuracy in the high 70s and an AUC of about 79.5% is consistent with what is typically seen for tree-based models on tabular clinical data of this size, and is notably more modest than some competing studies the authors cite (which report AUCs of 85–94%) — a good sign that the authors are not overstating their results. That said, several potential biases exist. The class imbalance (75% recovered vs. 25% died) can bias trees toward the majority class unless class weighting was applied, and the paper does not clearly state whether such weighting was used. The dataset is drawn from only two centres, so region-specific treatment protocols or population characteristics may not generalise to other healthcare settings. Complete-case analysis (dropping any patient with a missing value) could also systematically exclude the sickest or most complex patients, whose records are more likely to be incomplete. To strengthen the findings, the authors could report confidence intervals or perform statistical significance testing between the four trees' metrics, validate the model on an external hospital's data, and test the model's sensitivity to different missing-data handling strategies (e.g., imputation vs. deletion).

(c) Real-World Applicability

A model like this CART tree is well-suited for deployment as a bedside or triage decision-support tool in hospital emergency and ICU settings, helping clinicians rapidly flag high-risk patients using only six routinely collected features, without needing imaging or expensive tests. It could also inform resource allocation (ICU beds, ventilators) during surges in patient load. Scaling this to broader industry or national health-system use faces several challenges: data availability and standardisation vary widely across hospitals and countries, so the model would likely need retraining or recalibration on local populations; computational demands are low for tree inference, but the upstream data pipeline (lab integration, EHR connectivity) can be a significant engineering effort; and ethical issues arise around using an algorithmic mortality-risk score to influence triage or treatment prioritisation, which requires careful governance, clinician oversight, and transparency with patients and families.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

- Insight 1 — Interpretability vs. accuracy trade-off: This article reinforced that decision trees are valued in CO4 not only for their predictive power but for being "white-box" models. Compared with black-box ensembles or deep models covered under statistical learning, a single CART tree can be visualised as a flowchart that clinicians can directly follow, connecting to the course's emphasis on inductive learning as a transparent, rule-generating process.
- Insight 2 — Feature selection as a precursor to tree learning: Using a Random Forest purely to rank feature importance before fitting the final interpretable model illustrates how ensemble/statistical learning methods (CO4) can be combined rather than treated as competitors — an important nuance beyond simply comparing single algorithms.
- Insight 3 — Impurity measures drive tree structure: Seeing Gini index (CART) and information-gain ratio (C4.5/C5.0) applied to real clinical thresholds (e.g., $\text{BUN} \leq 20$) made the abstract concept of impurity-based splitting from the course concrete — the same entropy/Gini formulas taught in class directly determine which lab value becomes a decision node.

(b) Proposed Extension

A novel extension would be to combine this decision-tree framework with reinforcement learning for sequential treatment decisions rather than a single one-time mortality classification. Instead of predicting mortality risk once at admission, an RL agent could be trained to recommend a sequence of interventions (e.g., escalate monitoring, order additional labs, ICU transfer) as new lab results arrive over a patient's stay, using the interpretable CART splits as engineered state features so the policy remains partially explainable. This is feasible because the same hospital information systems already capture time-stamped lab values, and offline RL techniques (e.g., using historical patient trajectories as a fixed dataset) avoid the ethical problems of experimenting on live patients. The expected impact would be a decision-support system that adapts recommendations dynamically as a patient's condition evolves, rather than issuing only a static risk score, better matching how real clinical decision-making unfolds over time.